Design and development of the Web Based Nia Digital Museum: Software Requirements Specification - Linda Umwali, ALU, 25th January 2026

- **My mission**
  I believe in passion and talent. My aim is to shift mindsets from formal jobs to self employment. I want to inspire people to pursue their passion and use their natural abilities to get profit. My main focus will be art because art is important to Africa, for it tells our stories, shows our cultures and inspires new generations to keep it rich and spread.

- **Problem statement:**
  In Africa, unemployment is a major challenge especially among the African youth population due to lack of access to quality education to participate in a modern workforce. Aside from that, many employed individuals remain completely unhappy with their careers due to misalignment between their careers and personal interest but choose to stay just for the sake of money and survival. Reports show that 18% of workers are dissatisfied and dislike their jobs(Gallup, 2023). Most of the time is spent at work, so your career shouldn't make you feel miserable or worse, impact your mental health.

- **Software development model: Evolutionary prototyping**
  The requirements of this project are expected to change and improve over time, especially the design elements, the preservation and display of content, and the interactions between users. So, it is appropriate to apply an evolutionary prototyping method because it allows the system to evolve gradually while remaining usable at every stage of development.
  The platform's emphasis on artists' information such as artwork display, exhibitions and other features, makes it better not to set aspects at the beginning but leave room for continuous revision and application of feedback throughout the project instead of determining them at the beginning.
  How this model will be used:
  - Initial working prototype: a basic but functional version of the platform will be developed and the core features such as viewing artworks, browsing artists, and basic artist profiles will be implemented.
  - User feedback and evaluation: artists and visitors interact with the system, then feedback is collected on usability, layout, clarity, and missing features.
  - Improvement: The same prototype is continuously improved and new features are added gradually.
  - Refinement and stabilisation: As requirements become clearer, the system

becomes more stable, performance, security, and data management are improved.
    - Final deployment: The evolving system will lead us to the final product where the system reflects real user needs and refined choices.
- **The hypothesis of the solution**

  Once a workable version of the NIA Digital Museum is implemented, the vision is that the platform can greatly enhance visibility, access, and preservation of an artist's work, especially young and less represented creatives.

  It is expected that:

  - Artists will have an opportunity to showcase their talents in a digital space without having to rely solely on their physical gallery.

  - The visitors will enjoy the ease of accessing curated works of art and exhibitions across geographical locations.

  - This platform will facilitate a culture of appreciation that is equally aligned with the project's aim for purpose-driven creation.

  - The feedback obtained from users will provide validation for its efficacy as an alternative to physical museums.

Title <Software Requirements Specification for the Web Based Nia Digital Museum, Linda Umwali, ALU, 25th January 2026>

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<25th January 2026>

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Revision History

Name	Date	Reason For Changes	Version

Linda Umwali	19/1/2026	Made a copy of the template and made minor changes.	Version 1.0
Linda Umwali	22/1/2026	Read through the entire document, and added a few sections.	Version 1.0
Linda Umwali	24/1/2026	Added requirements, while working offline.	Version 1.1
Linda Umwali	25/1/2026	Added all pre-prepared ideas and added diagrams before submission.	Version 1.2

1. Introduction

1.1 Purpose

The purpose of this software requirement specification document is to define both functional and non-functional requirements for the Nia digital museum, version 1.0. This document will be clearly explaining the system's behaviour, constraints, and interfaces to guide the design and development process as well as stakeholders involved. This SRS document serves as a reference to help understand what the system will do, how it will behave, and who will use it. The core features this SRS will cover include user access, content viewing, and content management. Advanced features such as specific algorithms, code structures, sales analytics and project management plans are outside the scope of this document.

1.2 Document Conventions

This SRS document follows a structured format using numbered sections and headings for clarity. This document contains some prefixes including Functional requirements labeled using prefix FR, non-functional requirements are labeled using NFR, SRS stands for software requirement specification and BR is short for Business rule. Content is organised using bold headings , and tables are used to improve readability.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers, project supervisors or instructors, system administrators, testers and future maintainers of the system. Developers will use this document to understand the functionalities of the system as well as the constraints, while testers will rely on defined requirements to verify the system behavior. Project supervisors and instructors may use this document to evaluate the completeness and feasibility of the proposed solution.
This document is organized into sections that explain the system overview, requirements, and system constraints. Readers are encouraged to read starting with the introduction and description sections to gain an understanding of the project in whole. Developers and testers should focus on the FR and NFR sections, while administrators and maintainers may refer to sections describing system behavior and constraints.

1.4 Product Scope

The Artfrique digital museum system is a web-based platform designed to showcase, preserve, and provide access to digital artworks and cultural content. The purpose of this project is to help people showcase their talents through art pieces, and cultures through canvas regardless of location. This system will support artists to sell their products and gain spotlight, while offering users an engaging visual experience.
Key objectives of the system include expanding opportunities for artists, connecting them with a wide web audience through advertisement, creating immersive virtual art displays and promoting African cultures through art works and exhibitions. Features such as monetization, analytics, or complex project management functions are considered outside the scope of this version.

2. Overall Description

2.1 Product Perspective

The Nia digital museum is a brand new entity with its own self contained web environment that aims at providing solutions specifically for promotion and management of digital art works and other content within cultural domains. The system will not be an extension or a replacement for an already existing one but may have potential connections with other web systems or databases in the future implementation with artist portfolios or educational tools found in other internet sources.

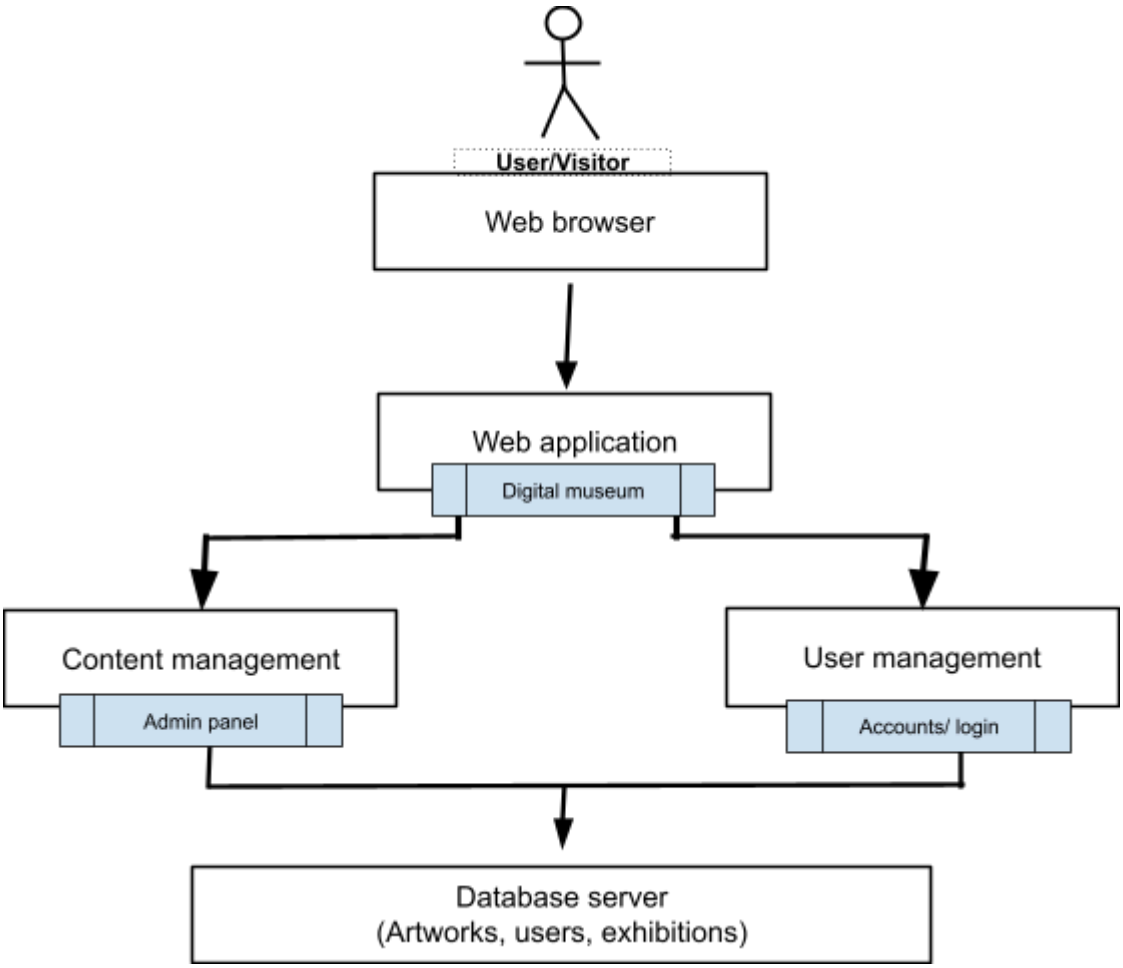

Fig1. Conceptual Diagram for Nia Digital Museum

2.2 Product Functions

- User login and access management
- Browse and view artwork exhibitions
- Search and filter artworks by artist, title, or, category
- Admin content management such as add, edit, delete artwork
- Organise artworks into collections
- Display details about all artists

These functions represent the core capabilities of the system and form the foundation of the detailed functional requirements described in section 3.

2.3 User Classes and Characteristics

User class	Description	Characteristics	Technical expertise
Visitor	Public users exploring the museum.	● Access only to view content ● no editing priviledges	Minimal, just for browsing
Registered user/artist	Content creators who upload and manage their artworks.	● Can add, edit, or delete their own content ● Can view analytics of their pieces ● Can manage profile	Basic to intermediate, able to use web forms, upload images, manage content.
Administrator	Museum staff or curators	● Full access to management ● Exhibitions ● Metadata	Advanced technical skills required, for maintaining content quality, organise exhibitions, manage users, ensure proper system operation.

The administrator is the most critical user class, while visitors represent the majority of system use.

2.4 Operating Environment

The Nia digital museum is a web-based application accessible via standard web browsers including Google Chrome, Microsoft Edge, or Firefox. The system will operate in PCs, laptops, tablets, mobile phones with internet connectivity. It requires a web server and relational database for storing content, and it must be compatible with common operating systems including windows, macOs, iOS,and android.

2.5 Design and Implementation Constraints

The system development will be done using web technologies like HTML, CSS, JavaScript, and a back-end scripting language like Node.js, with the application using a relational database for storage like MySQL. However, the constraints would include cross-browser compatibility, using

a device-friendly design, maintaining administrator-level authentication security, and web accessibility guidelines. Also, the libraries used would be well-maintained libraries by the respective API creators themselves.

2.6 User Documentation

The system will come with an online user documentation, including a user manual for visitors, and an administrator guide for content management too. Tutorials and walkthrough may be provided to help new users in navigating the platform.

2.7 Assumptions and Dependencies

The system assumes the user is able to access an internet-connected device, and any standard web browser. The system relies upon an online web server or database interface infrastructure. The libraries, which may be used for utilities like searching images, may be system compatible or otherwise maintained. Any changes to the browser or database would impact the system in any possible ways.

3. External Interface Requirements

3.1 User Interfaces(UI)

The NIA Digital Museum will offer a web-based graphical user interface, which can be accessed using any standard Internet web browsers. The graphical user interface will be visual, intuitive, and art-oriented, focusing on simplicity and aesthetics.

Core Interfaces in the system	
Visitor interface	Enables users to explore artworks, exhibitions, and artist profiles.
Artist Interface	facilitates artists in creating a profile, uploading their artwork, and monitoring their exhibition activities.
Administrator Interface	This interface enables system administrators to manage users, control content, and execute exhibitions.

Typical UI characteristics:

- Responsive design with support for desktops, tablets, and mobile devices.
- All pages have used consistent navigation and layout.
- Components that are available on all pages, e.g., navigation menus, search bar, login/logout options, available help.
- Clear visual feedback for user actions, such as success messages or errors.
- Error messages in an easily comprehensible format.
- The use of icons and visual previews for artworks to improve user experience.

3.2 Hardware Interfaces

The NIA Digital Museum does not require specialized hardware components. It will operate on standard computing devices including:

- Desktop and laptop computers
- Tablets
- Smartphones

3.3 Software Interfaces

The system will interface with several software components to operate effectively:

- Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
- OS: Windows, macOS, Linux, Android, iOS.
- Database Management System: A relational database-such as MySQL or PostgreSQL-that will be used to store user information, artwork metadata, exhibitions, and transactions.
- Web Server: This is any server environment, like Apache or Nginx, which can host an application.
- Backend Services: APIs for authentication, artwork management, and content retrieval.
- Cloud storage services for storing high-quality artwork files. This includes data such as user credentials, photos of artwork, their descriptions, prices, and exhibitions. All data exchanges shall be done in secure and standardized formats, for instance, JSON.

3.4 Communications Interfaces

The NIA Digital Museum will utilize standard protocols for internet communication.

Protocols for Communications:

- HTTP/HTTPS for web communication.
- HTTPS will be imposed to guarantee data transmission in a secured manner.

Network Communication:

- Client-server communication between the user's browser and the web server.
- Communication between the frontend and backend services using APIs.

Security & Encryption:

- SSL/TLS encryption for data in transit.
- Authentication and authorization mechanisms to protect the available actions.

Additional Communication Features:

- Email notifications while an account is created, an artwork gets approved, and a transaction is confirmed
- Online forms for submission of artwork, user registration, and feedback.

These communication interfaces will ensure interaction with users of the NIA Digital Museum system reliably, securely, and efficiently.

4. Requirement Specification

4.1 Functional Requirements

Req ID	**Requirements**	**Description**
FR 1	**User access**	The system must allow users to access the platform through web browsers.
FR 1.1	Visitor login	Visitors can browse content without registration.
FR 1.2	Registered user login	Registered users can login using username and password.
FR 1.3	Admin login	Administrators can login with secure credentials to access the admin panel.

FR 2	**Content viewing**	Users can view artworks, exhibitions, and cultural content.
FR 2.1	Search artworks	Users can search for artworks by artist, title, or category.
FR 2.2	Filter artworks	Users can filter artworks by type, style, or exhibition.
FR 2.3	View artwork details	Users can view detailed metadata including title, artist, description, and date.
FR 3	**Content management**	Administrators can manage all artworks and exhibitions.
FR 3.1	Add artwork	Admins can upload new artwork, including images, metadata, and category.
FR 3.2	edit artwork	Admins can edit existing details of artworks.
FR 3.3	Delete artwork	Admins can remove artworks from the platform.
FR 4	**User interactions**	Registered users can engage with content.
FR 4.1	Bookmark Artwork	Users can bookmark their favourite artworks.
FR 4.2	Comment on artwork	Users can leave comments on artworks
FR 5	**Notifications**	The system may notify users or admins about updates.
FR 5.1	Admin alerts	Admins are notified when content is submitted or uploaded.
FR 5.2	User Alerts	Users may receive updates about exhibitions or new artworks.
FR 6	**Art monetization**	The system must support artwork sales functionality.
FR 6.1	Set artwork price	Artists can assign prices to their artworks.
FR 6.2	View purchase details	Users can view artwork pricing and purchase information and status.
FR 6.3	Purchase request	Users can submit a request to purchase an artwork.
FR 7	**Exhibition management**	The system must allow the organisation of artworks into exhibitions.

FR 7.1	Create exhibition and assign artworks	Admin can create virtual exhibitions by grouping artworks by similarity or by features.
FR 7.2	View exhibitions	Users can browse exhibitions as curated collections.

4.2 Nonfunctional Requirements

Requirement Type	**Req ID**	**Description**
Security	NFR 1	● The system must authenticate users using secure login credentials ● Only authorised users like admins and artists can upload, edit, or delete artworks.
Performance	NFR 2	● The system must support at least 20 concurrent users without performance degradation. ● Artwork pages must load within 5 seconds under normal network conditions.
Usability	NFR 3	● The language used to communicate with users is English ● The user interface must be easy to use and require minimal training.
Reliability	NFR 4	● The system must ensure that none of the user data or artwork is lost. ● Data consistency: data must remain accurate and uniform.
Compatibility	NFR 5	● The system must be accessible across the two web browser ● Google Chrome ● Microsoft Edge
Maintainability	NFR 6	● The system must allow easy maintenance and future upgrades.
Technology	NFR 7	The system must be accessible on: ● PC ● Android ● iOS

4.2.1 Performance Requirements

The system must also be able to handle the requests as quickly as possible so as to provide a smooth browsing experience to users, more so when opening the images as well as the exhibitions.

4.2.2 Security Requirements

The system also needs to maintain user account security, artwork data, and artist content by ensuring authentication and access controls are implemented.

4.2.3 Usability Requirements

The system should be intuitive and easily understood at a visual level so that it may be easily navigated by the artist, visitor, and administrator.

4.2.4 Reliability and scalability

The system has to be stable during normal use and support future growth in terms of its content, user base, and features, not requiring changes in its architecture.

4.3 Software Quality Attributes

- Usability

The system should also be easy for non-technical end-users, including artists and visitors. The system will have clear navigation so that the end-users can easily understand and start browsing the system even without any sort of training.

- Reliability

The system shall be available a minimum of 95% of the time, not including scheduled maintenance, without any loss of uploaded artwork and user data.

- Maintainability

The system should be developed based on modular architecture to support possible future enhancement without affecting its current functionality.

- Scalability

The system shall facilitate an increase in the number of users, artworks, and exhibitions without significant degradation in system performance.

- Security

The system must also support role-based access control to guarantee that only authorized users can manage their own content and/or user accounts.

- Portability

The system shall be accessible for use on different platforms such as desktop machines, laptops, tablets, and smartphone devices using modern web browsers.

- Interoper

The system should be able to accommodate integration with external sources such as image storage tools in future versions of the software.

- Robust

The program must handle bad input or user actions properly by displaying useful error messages without terminating itself.

4.4 Business Rules for the Nia digital museum

BR 1: Only administrators have access to approve, alter, or delete any form of art or exhibition posted on the platform.

BR 2: Only the artists themselves can create, edit, or delete works which they themselves created.

BR 3: Non-registered users can view artwork, but they cannot upload, make comments on, or purchase artwork.

BR 4: Artists are required to create an account before they upload any artwork on the platform.

BR5: The pricing information for an artwork should be set by the artist and approved by an administrator before it becomes visible to users.

BR 6: All content uploaded must meet the Content Guidelines and Cultural Sensitivity policies set on the platform.

5. Appendix

This section provides other information that would supplement the core SRS but that does not form part of either the functional or non-functional requirements. They are important to the system for descriptive and explanatory purposes: technical, legal, and conceptual.

Additional requirements

- Database requirements

The system shall use a relational database to store user accounts, artist profiles, artworks, exhibitions, and metadata such as title, description, categories, and prices.

- Requirements of internationalization

The system should be designed with future multilingual capabilities in mind whereby the system can display content in several languages, one of them being French.

- Legal and Ethical Requirements

All artworks uploaded must not infringe on any copyrights. The ownership of the artworks remains with the artists, as this site is only a digital showcase and distribution medium.

- Reuse Objectives

The system components shall be designed to be reused in any future cultural, educational, or creative platform.

Appendix A: Glossary

Term	**Definition**
Digital Museum	A web-based platform used to present and preserve art and other cultural items.
Artwork	A digital representation of an artistic or cultural creation uploaded to the system.
Artist	A registered user who uploads and manages

	artworks on the platform.
Visitor	A user who browses the digital museum without content management privileges.
Administrator	A user responsible for managing users, content, and exhibitions.
Exhibition	A curated collection of artworks grouped for presentation.
SRS	Software Requirements Specification document.
UI	User Interface.
NFR	Non-Functional Requirement.
FR	Functional Requirement.
OS	Operating system

Appendix B: Analysis Models

The following are models which may be used to aid system understanding and further development:

1. Use case diagram

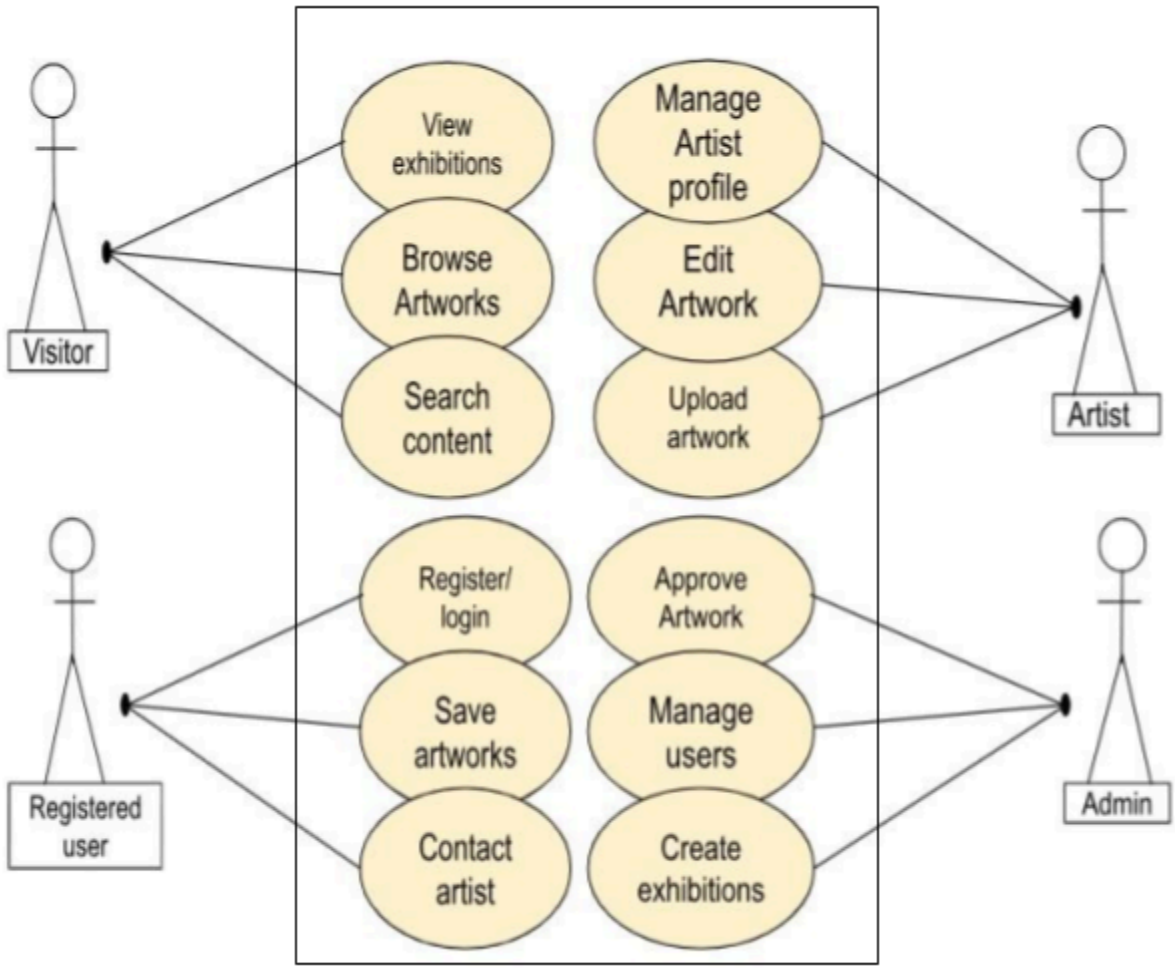

Fig2. Use Case Diagram.

This diagram shows the interaction between the users, which are the visitors, the artists, the administrators and the system functionalities.

2. Activity diagram - Artwork upload process

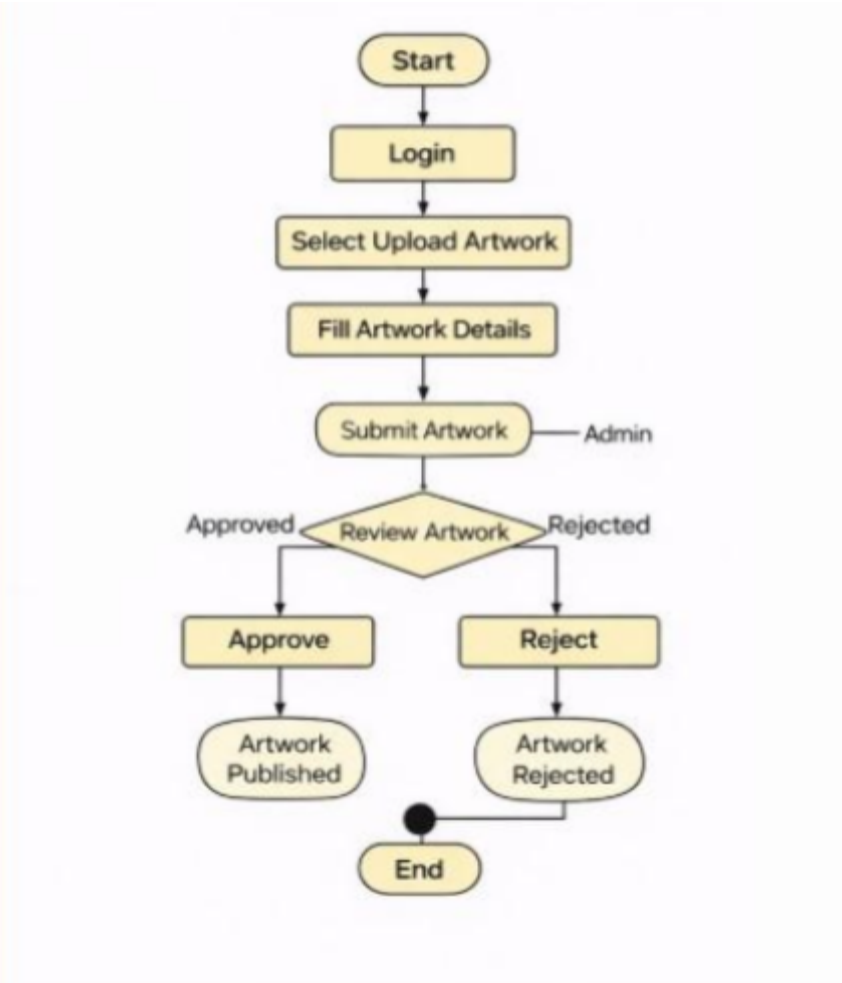

Fig3. Activity Diagram.

The Activity Diagram depicts the workflow of how an artwork is uploaded into the system. The sequence of actions from the login by the artist to the uploading of the artwork, and then the decision on whether to accept or reject it by the administrator, is depicted in the Activity Diagram.

3. Class diagram

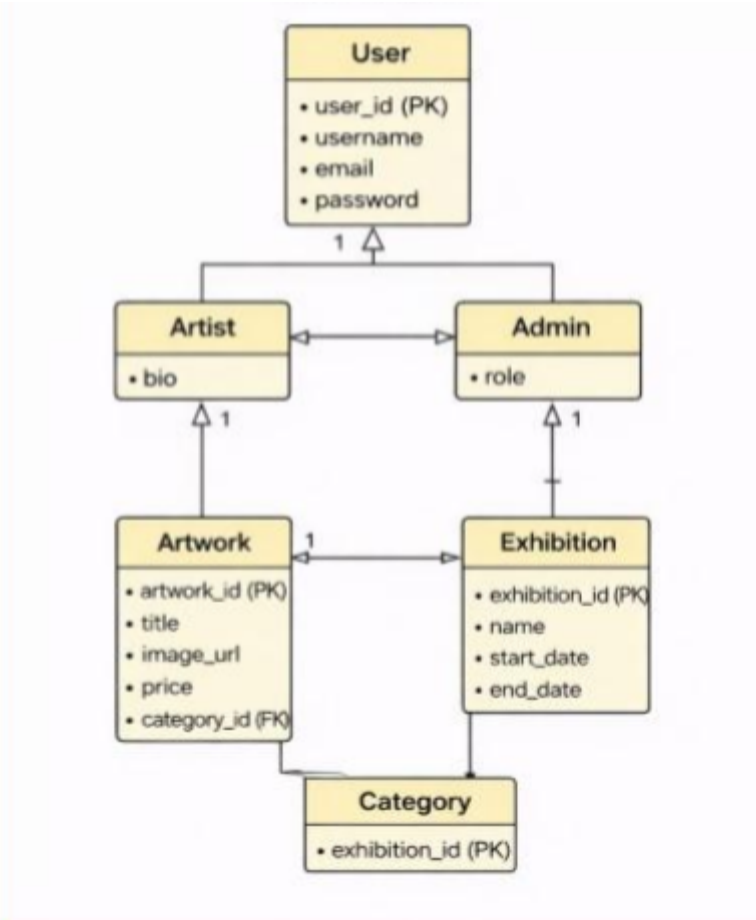

Fig4. Class Diagram.

The Class Diagram describes the static structure of the NIA Digital Museum system. The diagram includes the main classes, which are User, Artist, Admin, Artwork, Exhibition, and Category. The Class Diagram describes the relationship between the classes, including the inheritance relationship between the User class and the subclasses, which are Artist and Admin. It also describes the relationship between the classes, including the relationship between Artist and Artwork, and between Exhibition and Artworks.

4. Sequence diagram

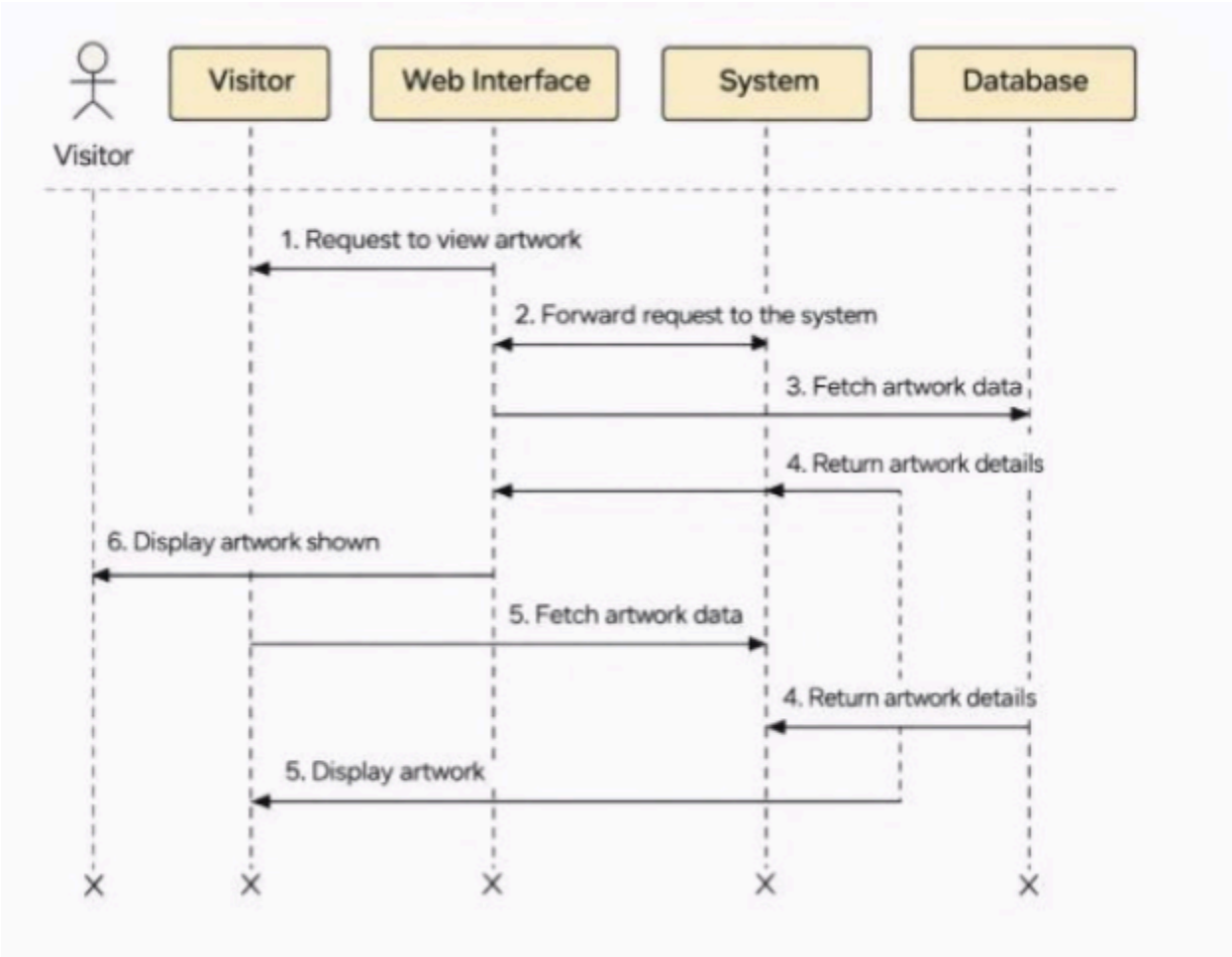

Fig5. Sequence Diagram.

As shown in the Sequence Diagram above, it represents the interaction between the different components of the system when the visitor wants to view an artwork.

OTHER DIAGRAMS

1. ERD

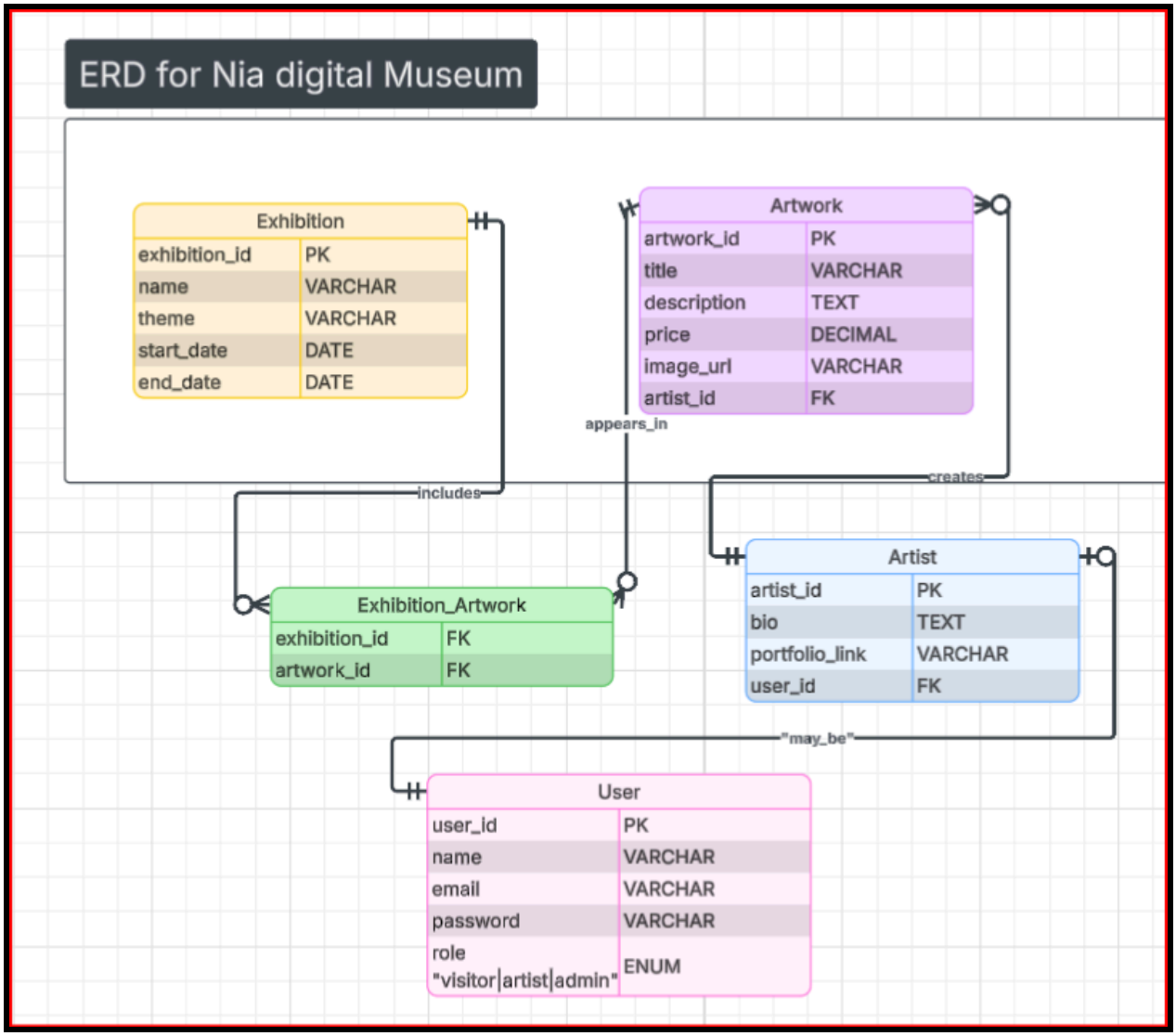

Fig6. Entity Relationship (ER) Diagram

This diagram represents relationships between entities such as Users, Artworks, Exhibitions, etc., within the system.

These models are purely conceptual and can be modified in the system designing and development phases.

6. REFERENCES

Gallup. (2023, July), *How is life at work going?*
https://www.gallup.com/workplace/508895/life-work-going.aspx